

TugOS

The Case for a Modern Tugboat Operations Platform

Market Analysis · Technology Gaps · Open Source Landscape · OSL Architecture

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The global tugboat and inland marine towing industry operates on razor-thin margins yet relies on decades-old record-keeping methods — paper logbooks, disconnected spreadsheets, and legacy desktop software that predates smartphones. This white paper identifies the operational pain points, surveys the competitive landscape (including the incumbent leader), examines available open-source foundations with their real license terms, and presents a strategic architecture for a purpose-built vertical SaaS platform that Open Scaffold Labs is positioned to deliver as the affordable, mobile-first alternative for small and mid-size operators.

\$1.2–\$2.4B	10–17.5%	Largest share: N. America	Legacy prevalence: HIGH
Marine mgmt. software market, 2024 (range across research firms)	Market CAGR estimates, 2025–2034 (range across research firms)	U.S. is the core beachhead geography	Widely reported; exact share unquantified — validated in Phase 1 discovery

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01 · The Tugboat Industry — Operational Context

Tugboats are the workhorses of maritime commerce — harbor tugs assist mega-vessels in and out of berths, inland tow boats push barge trains on river systems, and offshore salvage tugs respond to emergencies at sea. The U.S. alone operates thousands of commercial tugs and towboats, from single-boat owner-operators to fleets of 200+ vessels.

Key Operational Domains

- **Voyage & Job Dispatch** — assigning tugs to assist jobs, tracking ETA, communicating orders from port operations centers to captains.
- **Subchapter M Compliance** — since 2018, 46 CFR Subchapter M (Parts 136–144) is the central U.S. towing-vessel regime: every towing vessel 26 ft and over needs a Certificate of Inspection, obtained via USCG inspection or a Towing Safety Management System (TSMS) audited by a Third Party Organization. Record keeping for Sub M is the single largest compliance burden operators face — and the anchor requirement for any tug software product.

- **Crew Scheduling & Certification** — rotating crews on hitch schedules (7-on/7-off is common), tracking STCW endorsements and Merchant Mariner Credentials, and observing work-hour limits: licensed towing-vessel operators may not work more than 12 hours in any consecutive 24 (46 U.S.C. 8904), with rest-period rules in 46 CFR 15.1111.
- **Fuel Consumption Tracking** — diesel is 30–50% of operating expense. Accurate per-voyage fuel accounting is essential yet rarely achieved.
- **Maintenance & Class Surveys** — Sub M requires documented maintenance and inspection records; ABS and other class societies conduct periodic surveys requiring deep history.
- **Billing & Invoicing** — tug services bill by assist-hour, bollard pull rating, daily rate, or contract. Disputes are common when records are incomplete.
- **Financial Reporting** — multi-vessel operators need per-vessel P&L, cost-per-operating-hour, and fleet utilization — rarely available in real time.

02 · The Problem: Why Existing Systems Fail

2.1 The Paper & Spreadsheet Trap

Most small to mid-size operators (2–30 vessels) run on paper job tickets, Excel, and QuickBooks. The result: invoice leakage when paper logs are lost; certification expiry tracked in cells nobody watches; no per-voyage fuel feedback loop; crew double-bookings; incomplete records at TSMS audits and USCG inspections; and billing cycles stretching 30–60 days.

2.2 Enterprise Software Is Overbuilt and Overpriced

Platforms like ABS Nautical Systems, BASS, and Veson IMOS are built for deep-sea fleets of 50+ oceangoing vessels at \$50,000–\$500,000+ per year with implementation teams. A 10-tug harbor operator cannot justify that investment.

2.3 Road-Towing Apps Don't Translate

Towbook and similar products were built for road towing (motor-club billing, impound lots, roadside dispatch). The compliance frameworks, billing models, and crew regulations of marine towing are entirely different.

2.4 The Incumbent: Helm CONNECT

Helm Operations (Victoria, BC) is the purpose-built incumbent in workboat operations software. Helm CONNECT covers maintenance, compliance, personnel, dispatch and billing, reports 275+ companies on the platform, counts majors like Foss Maritime as reference customers, and markets Subchapter M compliance directly. Any honest strategy for this vertical starts from this fact: **the tug vertical has a leader — at the enterprise and mid-market end.** What Helm does not own is the long tail: thousands of 2–25 tug companies for whom Helm is too heavy, quote-priced, and office-centric. TugOS is designed for exactly that tail.

2.5 Pain Point Severity Matrix

Pain Point	Frequency	Revenue Impact	Compliance Risk	Priority
Invoice leakage from missing logs	Daily	High	Low	Critical
Sub M / TSMS record keeping & audit prep	Continuous	Medium	High	Critical
Manual crew certification tracking	Monthly	Medium	High	Critical
No real-time fuel cost data	Per voyage	High	Low	High

Pain Point	Frequency	Revenue Impact	Compliance Risk	Priority
Billing cycle delay (30–60 days)	Weekly	High	Low	High
No per-vessel P&L reporting	Monthly	Medium	Medium	Medium
Crew scheduling conflicts	Weekly	Medium	Low	Medium
No mobile access from vessel	Daily	Medium	Low	Medium

03 · Competitive Landscape — Commercial Solutions

The marine software market is fragmented by vessel type, company size, and geography. The landscape divides into five tiers — and unlike the conventional wisdom, the tug vertical does have a category leader.

Tier 0 — The Workboat Incumbent

- **Helm CONNECT (Helm Operations)** — purpose-built for workboats and tugs; maintenance, compliance (incl. Sub M), personnel, dispatch & billing; 275+ companies incl. Foss Maritime. Quote-based pricing, enterprise/mid-market orientation, office-first workflows. The benchmark TugOS must position against.

Tier 1 — Enterprise Ship Management ERPs

- **ABS Nautical Systems, BASS BASSnet, Veson IMOS, VoyageX AI, Shipmate ERP** — deep-sea fleet ERPs, \$100K+/yr, on-prem or heavy SaaS; wrong size and wrong workflows for harbor tug companies.

Tier 2 — Mid-Market Vessel Management

- **Seahub Fleet, SoftMar** (~\$1,200+/mo for 10 vessels), **DockMaster** (marinas), **Quadrup NORIS** — generic vessel management; none are tug-workflow specific.

Tier 3 — Inland Marine Specialists

- **BargeOps** — the closest inland specialist: towing, fleetings, crews, liquids, and analytics modules for barge lines; targets large river operators, priced and scoped beyond small harbor-tug companies. **OpenTug BargeOS** — VC-backed barge logistics. **Marine Job Planner** — Singapore port-authority job allocation.

Tier 4 — Road Towing (Adjacent, Not Applicable)

- **Towbook, TraxeroGO, OctopusPro** — road towing and field service; wrong domain entirely.

Key takeaway. The vertical has an incumbent (Helm CONNECT) at the top and a hard floor of paper/Excel at the bottom. What no one owns is the middle: a modern, mobile-first, self-serve, SMB-priced product for companies operating 2–25 tugs doing port assist, ship-docking, barge towing, or salvage work. That segment — thousands of U.S. companies — is TugOS's wedge, and the product must be built so a crew adopts it before an IT department ever gets involved.

04 · Open Source & Free Software Foundations

Open-source building blocks reduce development cost and time-to-market — but license terms matter for a commercial SaaS, and each component below is listed with its real license and an honest note on commercial usability.

- **Signal K** (Apache 2.0) — open marine data standard and server: GPS, AIS, engine, fuel flow via REST/WebSocket. Commercially safe. The backbone for vessel IoT in TugOS.

- **OpenCPN** (GPLv2) — mature chart plotter (current release 5.14.x, 2026). Useful for chart data and navigation primitives in the dispatch view; GPL obligations respected by keeping it as a separate process/data source, not linked code.
- **AIS data strategy** — the MarineTraffic AIS Toolbox (open-sourced June 2022) is licensed **CC BY-NC-SA 4.0 — non-commercial** — and is a density-map post-processing toolbox, not a real-time tracking stack. It is therefore **not** a foundation for TugOS. Real-time vessel positions come from commercial AIS APIs (MarineTraffic/Kpler, Spire, aisstream.io) and/or self-hosted AIS receivers with a permissively-licensed decoder (license-verify at selection).
- **Flectra ERP maritime module** (LGPL) — open-source ERP (Odoo fork) with maritime fleet, crew, and financial management. Candidate back-office layer; evaluate before committing.
- **STSS** (research/MIT, NUS Centre for Maritime Studies) — Python maritime traffic simulation using AIS data; useful for analytics and route optimization research.
- **OpenShipping.org** (open standard) — API standards for port and logistics integration.

05 · The Gap: What the Market Is Missing

Operator Type	Vessels	Current Options	What's Missing
Solo owner-operator	1–2	Paper / Excel / QuickBooks	Any digital system at all — even basic job tracking
Small tug company	3–10	Paper/Excel; road-towing apps hacked to fit	Maritime-compliant crew records, Sub M record keeping, tug-specific billing, self-serve pricing
Mid-size operator	10–30	Helm CONNECT (if budget allows), BargeOps, legacy Shipmate	Mobile-first UI, transparent SaaS pricing, fast deployment without implementation projects
Large tug fleet	30–100+	Helm CONNECT, enterprise ERPs	Served — not TugOS's entry market

The 3–30 vessel rows are the beachhead: real revenue, serious operational pain, and no product that is simultaneously tug-specific, affordable, self-serve, and crew-first. TugOS wins not by claiming the market is empty, but by being 10x easier to adopt than the incumbent for operators the incumbent was never designed for.

06 · TugOS — OSL's Proposed Vertical Platform

TugOS is a purpose-built, mobile-first SaaS platform for tugboat and inland marine towing operators — laser-focused on day-to-day tug workflows. Eight feature pillars:

Pillar 01: Dispatch & Job Board

- Real-time job board; drag-and-drop tug assignment with mobile push; GPS-tracked job progress (en route, on scene, assist complete, clear); VHF/voice memos attached to job records; port authority intake via OpenShipping-style APIs.

Pillar 02: Subchapter M & Crew Compliance

- The compliance pillar is built around **46 CFR Subchapter M**: digital TSMS record keeping (drills, inspections, maintenance, manning), TPO audit-ready binders, and one-tap generation of a vessel's Sub M record book for a boarding USCG inspector.
- Digital crew profiles — STCW endorsements, MMC, medical certs — with 90/30/7-day expiry alerts pushed to crew and management; hitch scheduling (7/7, 14/14, custom) with work-hour compliance against 46 U.S.C. 8904 and 46 CFR 15.1111.

Pillar 03: Fuel Tracking & Efficiency

- Per-voyage fuel logging (manual or Signal K IoT); cost-per-operating-hour dashboards; fleet fuel-efficiency benchmarking; bunkering records. Positioned as **direct cost savings** — tugs under 5,000 GT sit outside IMO CII scope, so the sell is economics, not regulation.

Pillar 04: Maintenance & Class Records

- Planned maintenance by calendar or engine hours; digital work orders with photos; Sub M and ABS inspection checklist templates; dry dock planning; parts inventory.

Pillar 05: Billing & Invoicing Engine

- Hourly assist / bollard pull / daily charter / contract models; auto-generated invoices from completed job records; client self-service portal; QuickBooks and Xero sync; disputed-invoice workflow with GPS track and log evidence attached.

Pillar 06: Financial Reporting & Fleet P&L

- Per-vessel revenue, cost, and margin; OPEX breakdown (fuel, wages, maintenance, port fees); utilization and idle time; budget vs. actual; custom report builder. *(Prototyped today in Vessel Finance.)*

Pillar 07: Mobile App (Captain & Crew)

- React Native app, iOS first; **offline-first with sync on reconnect** — non-negotiable for vessel connectivity; fuel/maintenance/job logging from the wheelhouse; digital signature capture at job completion.

Pillar 08: Integrations & Open Data

- Commercial AIS feed overlay on the dispatch map; OpenCPN chart data; Signal K sensors; NOAA weather and tides; port authority APIs where available.

07 · Technical Architecture & Stack

TugOS is built as App 24 on OSL's existing production architecture (the OSL Orchestrator Model — see the companion Architecture Blueprint v2). One stack, stated once:

Layer	Technology	Rationale
Web Frontend	React 19 + Vite + Tailwind	OSL component library and existing expertise; dispatch board and dashboards
Mobile App	React Native (Expo), iOS first	Shared logic with web; offline-first is critical on vessels
API Layer	Express.js on Vercel serverless	Same pattern as all OSL apps; REST + WebSocket for live job board
Database	Supabase PostgreSQL — dedicated TugOS project, RLS multi-tenant	Maritime clients and SOC 2 require a clean isolation story; TugOS runs in its own Supabase project rather than the shared OSL instance, with row-level security per tenant
Auth	JWT (15-min web tokens) + refresh-token grace for offline mobile	Roles: Fleet Admin, Port Captain, Dispatcher, Captain, Crew, Billing, Client; offline sessions must outlive short web tokens by design
IoT / Vessel Data	Signal K Server (Apache 2.0)	Fuel flow, engine hours, GPS without proprietary lock-in
AI / Intelligence	Paperclip (OSL) + Claude API	Fuel anomaly detection, predictive maintenance, auto-invoice drafting

Layer	Technology	Rationale
AIS / Tracking	Commercial AIS API (+ optional self-hosted receivers)	Real-time positions and ETA; NC-licensed open toolboxes excluded from the product
Billing	QuickBooks API + Stripe	Two-way accounting sync; card/ACH on client portal
Back-office (evaluate)	Flectra ERP maritime module (LGPL)	Possible accounting/reporting accelerator; decision after MVP

Multi-tenant from day one: each tug company is an isolated tenant enforced at the database layer. The dedicated-project deviation from OSL's shared-instance default is deliberate — it is the difference between passing and failing a port authority's security review.

08 · Go-to-Market Strategy

Positioning

“The modern alternative for working tug companies.” Helm CONNECT is the system enterprises buy; TugOS is the system crews adopt. Self-serve signup, transparent per-vessel pricing, live in a week, runs in the wheelhouse.

Pricing Model

Tier	Price	Includes
Deckhand (Free)	\$0/mo	1 vessel, 5 crew, basic job log, 30-day history — the zero-friction entry for solo operators
Harbor (SMB)	\$149/vessel/mo	Up to 10 vessels; full dispatch, Sub M records, crew compliance, fuel, billing, QuickBooks sync; minimum \$299/mo
Fleet (Growth)	\$99/vessel/mo	11–50 vessels; volume discount, priority support, custom reporting, API access, white-label portal
Enterprise	Custom	50+ vessels or port authority deployment; SLA, dedicated implementation, custom integrations

Acquisition Channels

- American Waterways Operators (AWO) — the primary U.S. towing trade association: conference presence, newsletter ads, Gulf Region chapter entry.
- Workboat magazine and maritime trade press; direct outreach via port dispatcher relationships; \$500 referral credits (captains talk to captains); content marketing — paper log vs. TugOS workflow videos, SEO for 'tugboat dispatch software' and 'Subchapter M software'.
- Free Deckhand tier as the funnel; conversion to Harbor as companies grow.

Beachhead Market

Gulf Coast first — Texas and Louisiana have the highest U.S. concentration of harbor tugs, inland towboats, and offshore support vessels. Sign 5 comped beta customers within 90 days of MVP; 20 paying companies by end of Year 1.

09 · Conclusion & Next Steps

The tug software market is not empty — it has an incumbent that owns the top and a paper trap that owns the bottom. The opportunity is the underserved middle: thousands of 2–25 tug companies with real revenue and no product built for how they actually work. OSL brings an existing production architecture, AI orchestration via

Paperclip, and open-source acceleration where licenses genuinely permit it. The wedge is focus: tug-specific workflows, Sub M-grade compliance, crew-first mobile UX, and pricing an owner can say yes to without a procurement process.

Discovery hypotheses (validate in weeks 1–2 before code)

- H1: A large majority of 2–25 tug operators still run on paper/Excel (the oft-quoted ~85% figure is unsourced — measure it in interviews).
- H2: Sub M record keeping and TPO audit prep is a top-3 pain and a purchase trigger.
- H3: Operators priced out of or overwhelmed by Helm CONNECT will pay ~\$149/vessel/mo for a lighter product.
- H4: A spot-assist job marketplace has real liquidity (harbor work is heavily contract/relationship-driven — do not build Phase 3’s marketplace until this tests true).

Timeline	Milestone	Description
Week 1–2	Customer Discovery	10–15 structured interviews across operator sizes; validate the hypotheses above; pick the three highest-priority MVP features
Week 3–4	Technical Spike	Signal K integration PoC; dedicated Supabase multi-tenant schema; basic dispatch board; commercial AIS feed evaluation
Month 2	MVP Build	Dispatch, job logging, basic crew records, invoice generation; internal alpha with one friendly operator
Month 3	Beta Launch	5 Gulf Coast beta customers; Deckhand tier live
Month 4–6	Commercial Launch	Harbor tier priced and marketed; AWO presence; 20 paying customers
Month 6–12	Scale & Expand	Fleet tier, port authority integrations, Series A data room if traction warrants

10 · The Best-in-Class Gameplan — Achieving Industry Leadership

Launching TugOS is the starting gun, not the finish line. The first company to build genuine operational trust with operators and crews in the SMB segment will define its standard. Four phases across 36 months:

Phase 1 · Months 1–6 · Build Trust on the Water

- Customer discovery before code — 15+ structured interviews; every MVP feature maps to a specific interview quote.
- **Hire or partner with a maritime domain expert** — a retired port captain or operations manager who has stood a watch. The Catalog of Authority (regulatory data connections) complements but does not replace a human who carries credibility at AWO conferences and in sales calls.
- MVP nails one workflow: digital job dispatch with crew notification and time-stamped logging. Offline-first from day 1 — if the app loses data when LTE drops, captains never trust it.
- 5 free Gulf Coast betas in exchange for weekly feedback, usage data, and case studies.

Targets: 5 beta customers · 15+ interviews · 500+ jobs logged · 0 data-loss events.

Phase 2 · Months 6–18 · Become the Operator's Operating System

- Launch the Sub M/TSMS compliance module — operators who keep years of compliance records in TugOS never leave. Cert tracking with 90/30/7 alerts; manning compliance reports.
- Automate billing from job records — two clicks from completed job to invoice with GPS track and crew list attached; this alone recovers more than the subscription cost.
- Captain app with signature capture; fleet analytics dashboard (utilization, revenue per operating hour, fuel trends — data most operators have never seen visualized).

- SOC 2 Type I in parallel (Vanta/Drata) — made credible by the dedicated-project tenant isolation; one port authority reference customer (target the 3rd–4th largest Gulf port).

Targets: 20+ paying companies · \$40K+ MRR · 200+ crew profiles · <5% monthly churn.

Phase 3 · Months 18–30 · Win the Category

- AI predictive maintenance on 18+ months of accumulated fleet data (Paperclip + Claude) — a data moat competitors cannot purchase.
- Annual 'State of Tugboat Operations' benchmark report from anonymized fleet data — positions TugOS as the industry's data authority.
- Spot Job Board marketplace (2–5% take) — **gated on hypothesis H4**: build only if Phase 1–2 evidence shows real spot-market liquidity.
- Public API and integration partners (payroll, insurance, bridge systems); international expansion (IMO/STCW frameworks travel well; multi-currency; Spanish and Bahasa).
- Series A at 75+ customers / \$150K+ MRR with the data moat as the story.

Targets: 75+ paying companies · \$150K+ MRR · 1,000+ vessel-months of data · 3+ countries.

Phase 4 · Months 30–36+ · Best in Class — What Separates OSL

- **The data moat** — every vessel-month of fuel, maintenance, and compliance history compounds; later entrants face a 2–3 year deficit.
- **Obsessive crew UX** — if a 55-year-old captain can't use it in the wheelhouse without training, it fails, regardless of how clever it is.
- **Compliance as the system** — when a USCG inspector boards, the captain pulls out an iPad and produces the Sub M record book in under 60 seconds. That moment is the best sales tool TugOS will ever have.
- **Community & trust** — AWO and Workboat every year, real customer stories with real numbers, a user community where operators swap practices.
- **Relentless iteration** — public roadmap, quarterly customer advisory board, 24-hour bug response SLA for paying customers.

The Competitive Separation Summary

Dimension	Incumbent / later entrants	TugOS
Target buyer	Enterprise & mid-market fleets, office-led purchase	2–25 tug companies, crew-led adoption, self-serve
Pricing	Quote-based, implementation projects	Transparent per-vessel SaaS, live in a week
Compliance	A compliance module	The Sub M record system used during actual USCG inspections
Crew adoption	Office-friendly UI	Wheelhouse-tested offline-first mobile UX
Data & AI	Dashboards	Predictive maintenance and fuel anomaly detection that compound with every vessel-month
Network effects	None	Benchmark reports; marketplace if and when liquidity is proven